Bayesian Analysis:
Modeling House Prices in the English Home Counties

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1 Introduction

House prices is a topic commonly used for modeling. They can be modeled using a simple linear regression, provided the data satisfies the underlying assumptions. There are some situations, however, when the dataset becomes more complex and we need to think about using some different, more suitable techniques. One example is data that includes hierarchical information about location. In such cases, observations are no longer independent (e.g., houses in the same area), and the hierarchical structure significantly increases model complexity and the number of parameters.

This is where Bayesian Hierarchical Models (BHMs) come in handy. Working with a dataset of properties sold in 2025 in England, we will use a hierarchical location structure, combining it with other property variables to analyze which factors (especially those related to location) are the most impactful. And since BHMs are conceptually similar to a frequentist Mixed Effects Models (MEMs), we will also compare the results and interpretations of these two approaches. Finally, we will evaluate the specific advantages of adopting a Bayesian framework in this context and see why it may offer superior insights for this type of spatial analysis.

2 Data

The data used in this project comes from Price Paid Data - an official UK website that contains the sale prices of properties in England and Wales submitted to HM Land Registry. The variables included in the dataset describe the exact sale price, date of transfer, property type (detached, semi-detached, terraced, flat or other), tenure (freehold/leasehold), and whether it is a new build. Apart from these categorical variables, we also have extensive information about the location: from counties and districts down to towns and streets. The data is available for download as monthly or yearly, and the period we chose for the analysis is the last complete year - 2025.

The full dataset about the houses sold in 2025 for England and Wales includes almost 800,000 records. This is a very large amount, especially considering there are over 300 districts in this region, and thousands of towns and streets. While it would be ideal to include every record and hierarchical level in the model, the computational requirements make this difficult, so we will work on a specific subset.

Firstly, we have decided to focus our analysis on the *Home Counties* (Figure 1) — the English counties immediately surrounding London. These traditionally include Surrey, Kent, Essex, Hertfordshire, Buckinghamshire, Berkshire, and West and East Sussex. However, as administrative divisions in England are often complex, several of these ceremonial counties are subdivided into smaller, independent Unitary Authorities.

In our dataset, this results in a total of 19 distinct administrative areas. Specifically, while counties like Surrey or Hertfordshire remain as single entities, others are split: *Berkshire* is represented by its six

unitary authorities (Reading, Wokingham, Bracknell Forest, West Berkshire, Slough, and Windsor and Maidenhead). *Essex* includes Southend-on-Sea and Thurrock, *Kent* includes Medway, *Buckinghamshire* includes Milton Keynes and *East Sussex* is paired with Brighton and Hove. A complete list of these units and their parent counties, based on the official classification by the Office for National Statistics (ONS), is provided in the Appendix B.

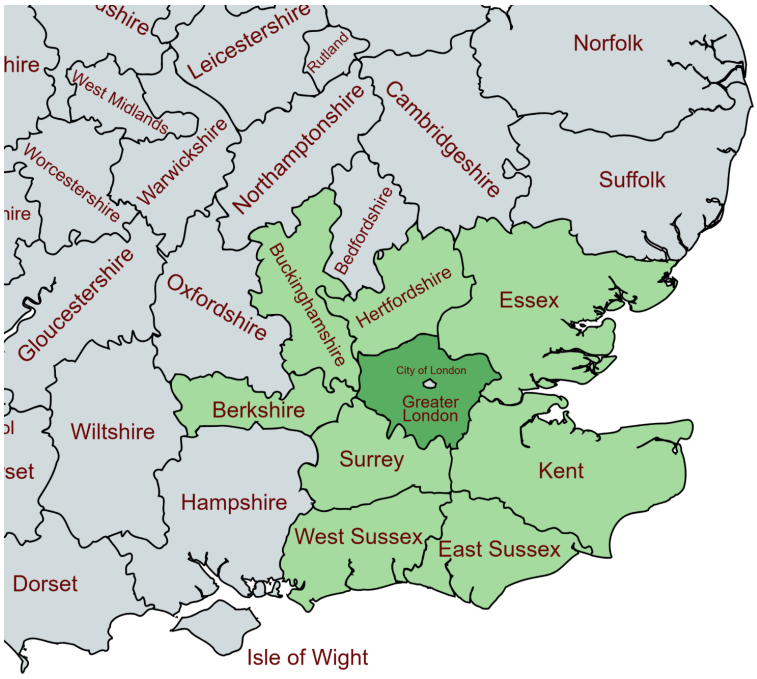


Figure 1: English Home Counties

Although focusing on these 19 counties reduced the dataset's scope, the remaining 200,000 observations posed a significant computational challenge for the BHM. To keep the model manageable while making sure the results remain reliable, we decided to work on a representative sample of the data. We chose a random 10% of the whole dataset, giving us a representative sample of 20,000 rows. In doing so, we ensured that even the smaller administrative units had enough observations for the model to perform reliably. This still reflects the reality of house sales in this area, while making the model computationally feasible.

At the end, limiting the scope in this way makes it possible to create a hierarchical model and allows us to answer interesting questions: how do the Home Counties differ in terms of price, and where are people relocating from London paying the most, and for what types of properties?

3 Model

The model used in the paper is a Bayesian Hierarchical Model. It is a multilevel model, which is used when the data are grouped into distinct categories - in our case those categories are the 19 counties in which the properties are located. An extreme approach in modeling this type of data would be to completely pool all the data and estimate a common vector of regression coefficients - but in that way

all the categories' individualities would disappear. At the other extreme, an approach with no pooling assigns each level its own coefficient vector, that is estimated separately from the other levels. This would give us separate models for each county, and an approach like this could increase the risk of overfitting and also make it difficult to compare results across different areas. A hierarchical model is an intermediate solution where the degree of pooling is determined by the data and a prior on the amount of pooling. [4].

The core idea behind BHM is to estimate group-level parameters by leveraging information shared across different, related groups. This allows the model to 'borrow strength' from the entire dataset, which improves the reliability of estimates for smaller categories where data might be sparse.

Each observation from our dataset is nested within a specific county, for which the model estimates a unique vector of coefficients. This way we get an estimation of varying intercepts and varying slopes, which capture both the baseline price differences between counties (the intercepts) and the localized impact of property features on market value (the slopes).

While this might seem similar to a 'no-pooling' approach, the fundamental difference lies in the treatment of groups - rather than being viewed as independent entities, the counties are linked through a shared probability distribution - a shared population-level prior that is itself estimated from the data. The variance of this prior (τ) determines the amount of pooling. If the data in each level are similar, strong pooling will be reflected in low hierarchical variance. If the data in the levels are dissimilar, weaker pooling will be reflected in higher hierarchical variance. [4]

To create the BHMs we used Stan, specifically via `rstan` library in R (code is provided in the Appendix A). Stan is a tool used to develop Bayesian models using Markov Chain Monte Carlo (MCMC) methods, specifically Hamiltonian Monte Carlo. The implementation of the model followed the approach outlined in [3]. We set the population-level regression coefficients (γ), which represent the global average effect, and the group-level coefficients (β), where the intercept and slopes are different for every county. The degree of this variation is determined by the hierarchical scale parameter (τ).

For every parameter, we need to set a prior distribution. Since we do not have specific prior knowledge, we used weakly informative priors. This way we allow the data to dominate the posterior distributions, but we are still providing enough guidance to stabilize the estimation process. For γ (the global effects), we set $Normal(0, 5)$. For τ, which describes the variability between counties, we used $Cauchy(0, 2.5)$ - as its heavy tails allow for larger variations between levels. Finally, for the residual error (σ), we set an $Exponential(1)$ prior, for strictly positive parameters.

To improve sampling efficiency and avoid divergent transitions, we implemented the model using non-centered parametrization. By reparameterizing the county-level effects as a scaled and shifted transformation of a standard normal distribution, it was possible for the NUTS sampler to more effectively explore the posterior distribution, particularly in areas where data for specific counties were sparse.

4 Results

The model was estimated using two independent Markov chains, each consisting of 2,000 iterations. All parameters achieved an $\hat{R}$ value of 1.0, confirming convergence and ensuring that the posterior distributions are reliable for inference, so now we can proceed to visualize the posterior estimates and explore the hierarchical dynamics of the Home Counties housing market.

4.1 Baseline Price Variation (Intercepts)

Figure 2 illustrates the distribution of property prices across the Home Counties (for a reference type of property: a detached, freehold, new building). Since the dependent variable was modelled as the logarithm of price, the intervals are asymmetric after back-transformation to the original scale. The dot represents the median for each county, while the thick line indicates the 66% (1σ) probability interval and the thin line shows the 95% (2σ) interval.

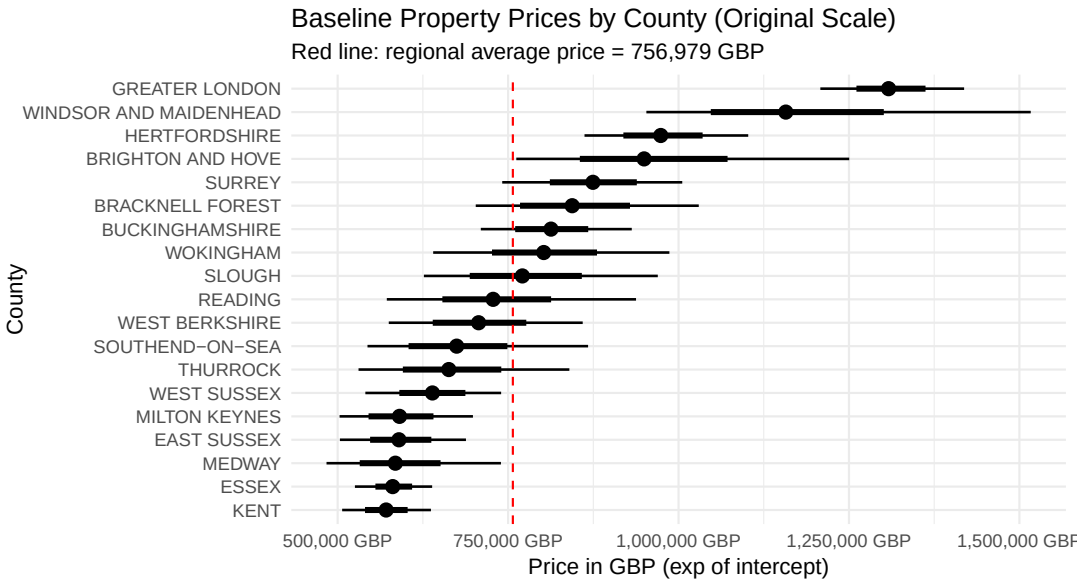


Figure 2: Average property prices in Home Counties

In general, Greater London stands out as significantly more expensive than the other regions. Its intervals are much shorter due to the high volume of data available for this area, leading to greater precision. On the other hand, the regions with the lowest baseline prices are Kent and Essex.

We can also observe that counties with wider intervals tend to be positioned closer to the regional average. These are the smallest regions with fewer observations, resulting in higher uncertainty. Their placement near the average is a result of shrinkage - a key mechanism of our hierarchical model. When the data for a specific category is limited, the model incorporates knowledge from the entire population to provide a more stable estimate.

4.2 Hierarchical Heterogeneity (τ)

We now move to the core component of the hierarchical model, which differentiates the impact of predictors across counties through both random intercepts and random slopes. The hyper-parameter τ represents the between-county standard deviation for each effect - a higher τ indicates greater geographic heterogeneity in how a specific variable influences property prices.

As shown in Figure 3, the Intercept has the highest degree of variation. This aligns with our observations in Figure 2 - that, for instance, a price of a house in a Greater London is very different from a price of a House in Kent. Furthermore, the variable "Property type: Other" (relative to the baseline of detached

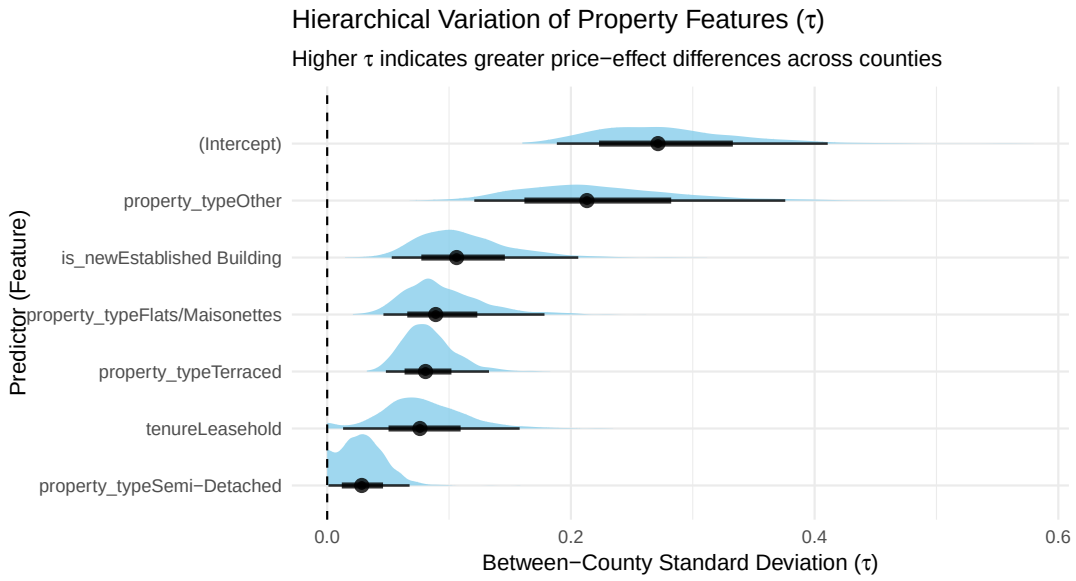


Figure 3: Standard variation in Home Counties for the house features

houses) shows significant variation, which can suggest that the value of non-standard properties is highly sensitive to local market conditions.

It is also important to consider the shape of the posterior distributions for τ. While the Intercept and "Property type: Other" have high mean estimates, their credible intervals are notably wide. This indicates that while the model identifies significant variability across counties, there is higher uncertainty regarding its exact magnitude.

In contrast, the "Semi-detached" category shows very low hierarchical variation. The posterior distribution for this variable is characterized by a high, narrow peak, suggesting the model is highly certain that the impact of being a semi-detached house is relatively uniform across all Home Counties.

4.3 Regional Slopes for Specific Features

Beyond the general review of hierarchical variability, we can examine specific county-level effects for individual predictors, similar to our analysis of the intercepts.

We begin with the feature having the second-highest degree of variation: the "Other" property type (Figure 4). The global average impact (γ_5) is estimated at approximately -0.42 on the log scale. This implies that, at a regional level, an "Other" property is approximately 34% cheaper ($e^{-0.42} - 1 \approx -0.34$) than a detached house, which is our base category.

As illustrated in Figure 4, Buckinghamshire exhibits the most negative effect. Its local coefficient of approximately -0.8 suggests that, in this specific market, such properties trade at a discount of roughly 55% compared to detached houses. Based on the 95% credible intervals, we can state there is a 95% probability that this parameter lies between approximately -0.95 and -0.65. In practical terms, this reflects a price discount ranging from 48% to 61%, an effect substantially stronger than the regional average.

In contrast, the effect in Milton Keynes is near zero, and its 95% credible interval overlaps with the zero-effect line. This implies that in Milton Keynes, the "Other" property type does not experience a

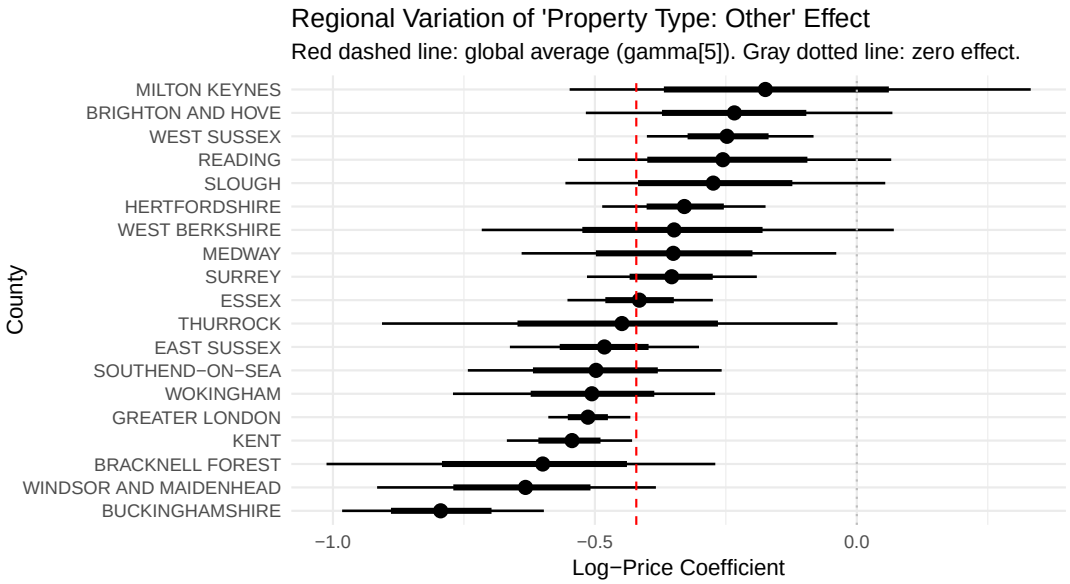


Figure 4: Regional variation of 'Property Type: Other' effect in Home Counties

statistically significant price penalty relative to the baseline. This suggests that the prices of non-standard properties in Milton Keynes are statistically indistinguishable from the ones of detached houses, marking it much different than Buckinghamshire, the example we analyzed before.

Next, we examine the distribution of parameters related to the difference between established buildings and new builds across various counties. In this case, the credible intervals are considerably wider for most regions, which results in estimates being pulled closer to the global average via the shrinkage mechanism. Notably, for the majority of counties, the 95% credible interval overlaps with zero. This indicates that, for those specific areas, there is no statistically significant price difference between established and new-build properties within this model's context.

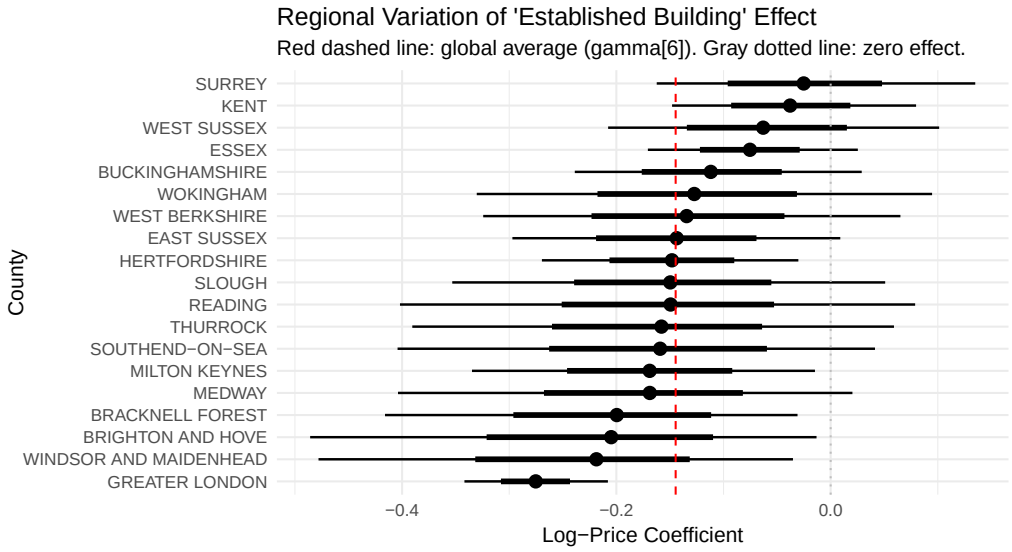


Figure 5: Regional variation of 'Established building' effect in Home Counties

The primary exception to this trend is Greater London, where the higher volume of observations leads to much narrower intervals and greater reaching certainty. For this county, we are 95% confident that the parameter for an "Established Building" lies between approximately -0.33 and -0.21. In practical terms, this means that an established property in Greater London is between 19% and 28% cheaper than a new-build equivalent ($e^{-0.21} - 1 \approx -0.19$ and $e^{-0.33} - 1 \approx -0.28$).

5 Comparison to the Frequentist Approach

To further validate the findings, we compare the Bayesian hierarchical results with a more classical frequentist approach: the Linear Mixed-Effects Model (MEM). Given our data structure and a previously created BHM, we specified a model with both random intercepts and random slopes across counties.

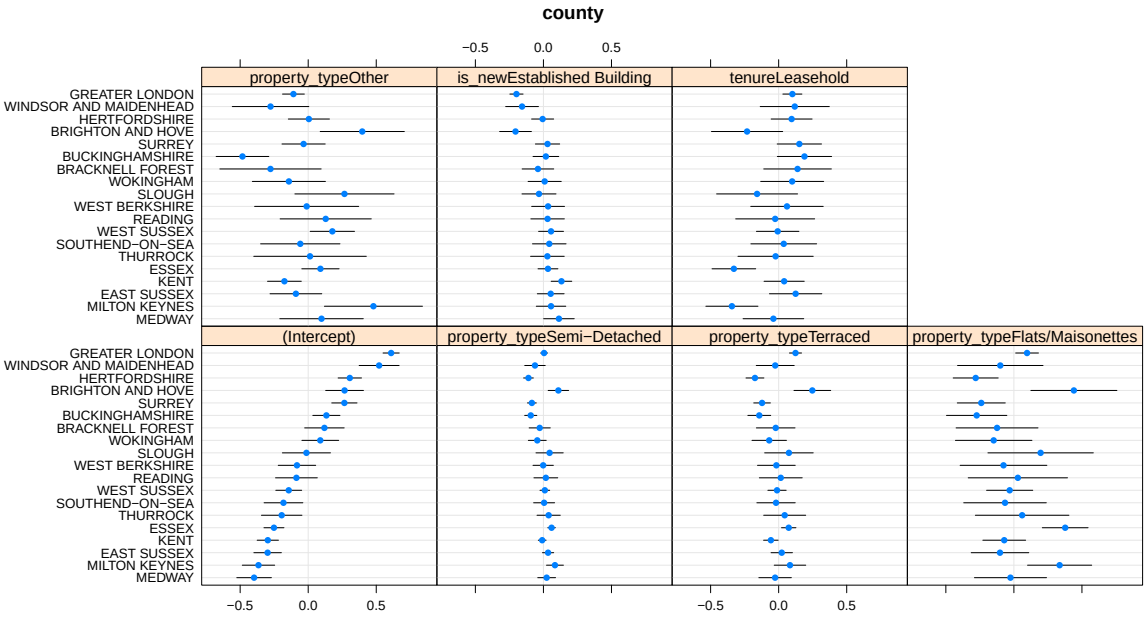


Figure 6: County-level coefficient estimates and 95% CIs (Frequentist MEM)

Overall, the results (Figure 6) are consistent with the previous approach. The ordering of counties by their effect sizes is mostly unchanged (for instance when we look at the intercept). This similarity is expected: Mixed-Effects Models also use a shrinking technique in estimation process [1]. Furthermore, because we utilized relatively weak priors in our Bayesian model, the posterior distributions were primarily driven by the data, leading to estimates close to the Maximum Likelihood results of the MEM.

However, key differences emerge in the quantification of uncertainty. As shown in Figure 6, while the general trends remain - such as Greater London having the highest intercept and most precise estimate, and Milton Keynes having the strongest effect for the 'Other' property type - the frequentist confidence intervals are significantly narrower than the Bayesian credible intervals.

This disparity also visible in the transformed average prices (Figure 7). But apart from the width of the intervals, the fundamental difference lies in their interpretation. In our Bayesian approach, we can state there is a 95% probability that the true parameter falls within a specific range. In the frequentist approach, a 95% confidence interval means that if we were to repeat the sampling experiment an infinite

number of times, 95% of such calculated intervals would contain the true parameter. The Bayesian interpretation is far more intuitive for the real-world decision-making and analysis of property prices than this repetitive sampling logic.

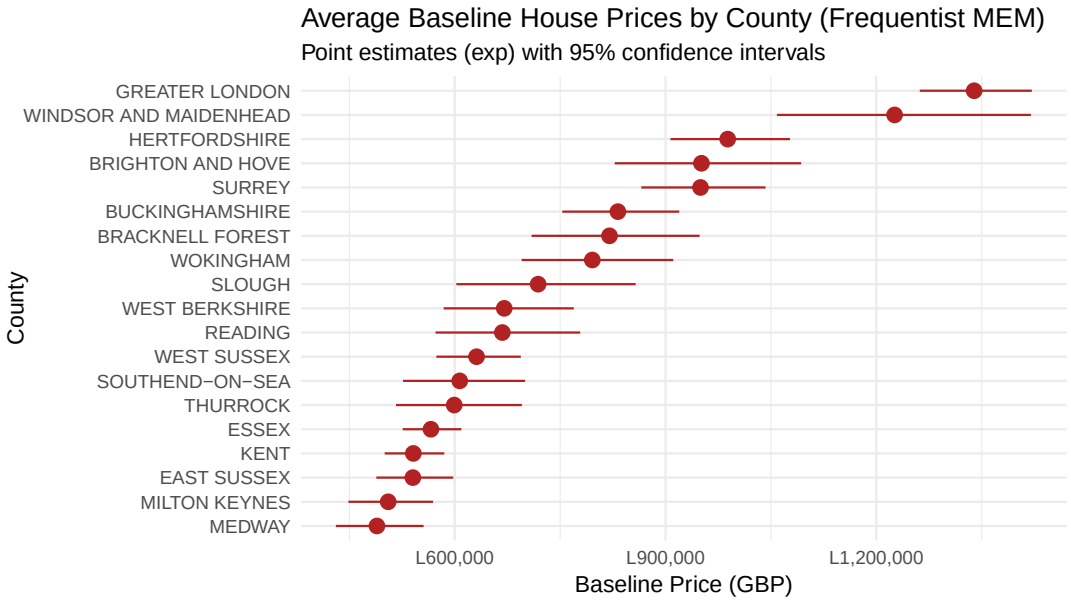


Figure 7: Average Baseline House Prices by County (Frequentist MEM)

One more important thing in the frequentist approach is the critical technical observation that occurred during the estimation of the MEM: the model produced a "boundary (singular) fit" warning. This indicates that for certain slopes or intercepts, the model failed to detect sufficient variation between counties, estimating some dimensions of the variance-covariance matrix as exactly zero. We can see those "collapsed" estimates in Figure 6, where several coefficients appear non-significant or identical across groups.

As described in the official documentation of the `lme4` package (used to create MEM), while singular models are mathematically defined, they can have several risks:

1. They may represent overfitted models with poor predictive power.
2. They are prone to numerical instability and convergence failures.
3. Standard inferential tools may become unreliable.

The primary solution to these issues is the adoption of Bayesian methods. By using a fully Bayesian framework (Stan), we regularize the model via priors that prevent the variance-covariance matrices from collapsing into singularity. Furthermore, the Bayesian approach provides credible intervals that formally account for the uncertainty in the random effects parameters - uncertainty that the frequentist MEM often underestimates or ignores when a singular fit occurs.

6 Concluding Remarks

In summary, we successfully implemented a Bayesian Hierarchical Model to analyze the housing market across the Home Counties in England. By employing this framework, we were able to establish baseline price estimates for each county while also capture how the impact of specific property characteristics, such as house type, age, and tenure, varies geographically. This together provides a powerful tool for real estate analysis, and offers a more nuanced understanding of local market dynamics than standard global regression models.

While a frequentist Mixed-Effects Model was estimated for comparison, the Bayesian approach turned out to be superior in several key aspects. Beyond its more intuitive probabilistic interpretation of credible intervals, the Bayesian framework remained numerically stable where the frequentist estimation struggled with convergence and singular fits. The use of regularizing priors allowed for the inclusion of complex random slope structures, ensuring that the model remained sensitive to regional heterogeneity without collapsing into a simplified, uniform solution.

It is also true, however, that the implementation of such a high-dimensional model presented significant computational challenges. Because of the large volume of data and the complexity of the MCMC sampling process, model estimation was very time consuming. Future research could focus on further optimizing the Stan code or using some higher performance computing machines to handle larger datasets. Such increased computational power would allow for an even deeper hierarchical structure - for instance, incorporating smaller administrative districts, postcodes, or even specific street-level data. Expanding the model in this direction would further enhance the precision of the estimates, and it could provide an even more granular lens through which to observe the complexities of the regional housing market.

References

[1] Michael Clark. Shrinkage in Mixed Effects Models.

[2] Office for National Statistics. United Kingdom: Local Authority Districts, Counties and Unitary Authorities, 2009. Official Map, ONS Geography, 2009. Accessed: 2026-04-27.

[3] R-bloggers. Hierarchical Models with RStan (Part 1), 2016. Accessed: 2026-04-20.

[4] Stan Development Team. *Stan User's Guide Version 2.22*, 2020. Accessed: 2026-04-20.

Appendix A Stan Model Code

The following code implements the Bayesian hierarchical model used in this study. It employs a non-centered parameterization for the random effects to improve MCMC sampling efficiency.

```
data {
  int<lower=1> N;                 // Number of observations
  int<lower=1> J;                 // Number of counties (19)
  int<lower=1> K;                 // Number of features (columns in matrix X)
  int<lower=1,upper=J> id[N];     // County indices for each observation
  matrix[N, K] X;                 // Predictor matrix
  vector[N] y;                    // Log-transformed house prices
}

```

```
parameters {
  vector[K] gamma;                // Population-level mean effects
  vector<lower=0>[K] tau;         // Between-county variability for each feature
  vector[K] beta_raw[J];          // Raw county-level deviations
  real<lower=0> sigma;            // Residual standard deviation (error term)
}

transformed parameters {
  vector[K] beta[J];              // County-specific coefficients
  for(j in 1:J){
    // Non-centered parametrization
    // Combines population mean, scale, and raw deviations
    beta[j] = gamma + tau .* beta_raw[j];
  }
}

model {
  vector[N] mu;

  // Weakly informative priors
  gamma ~ normal(0, 5);
  tau ~ cauchy(0, 2.5);
  sigma ~ exponential(1);

  for(j in 1:J){
    beta_raw[j] ~ normal(0, 1);  // For non-centered parameterization
  }

  // Linear predictor calculation
  for(n in 1:N){
    // Multiply property features by coefficients of the corresponding county
    mu[n] = X[n] * beta[id[n]];
  }

  // Likelihood function
  y ~ normal(mu, sigma);
}
```

Listing 1: Hierarchical Linear Model with Random Intercepts and Slopes in Stan

Appendix B Administrative Geography Map

This appendix presents the official administrative map that illustrates the division of England into Local Authority Districts, Counties, and Unitary Authorities, highlighting the structure within which the *Home Counties* are situated.

United Kingdom: Local Authority Districts, Counties and Unitary Authorities,[1] 2009

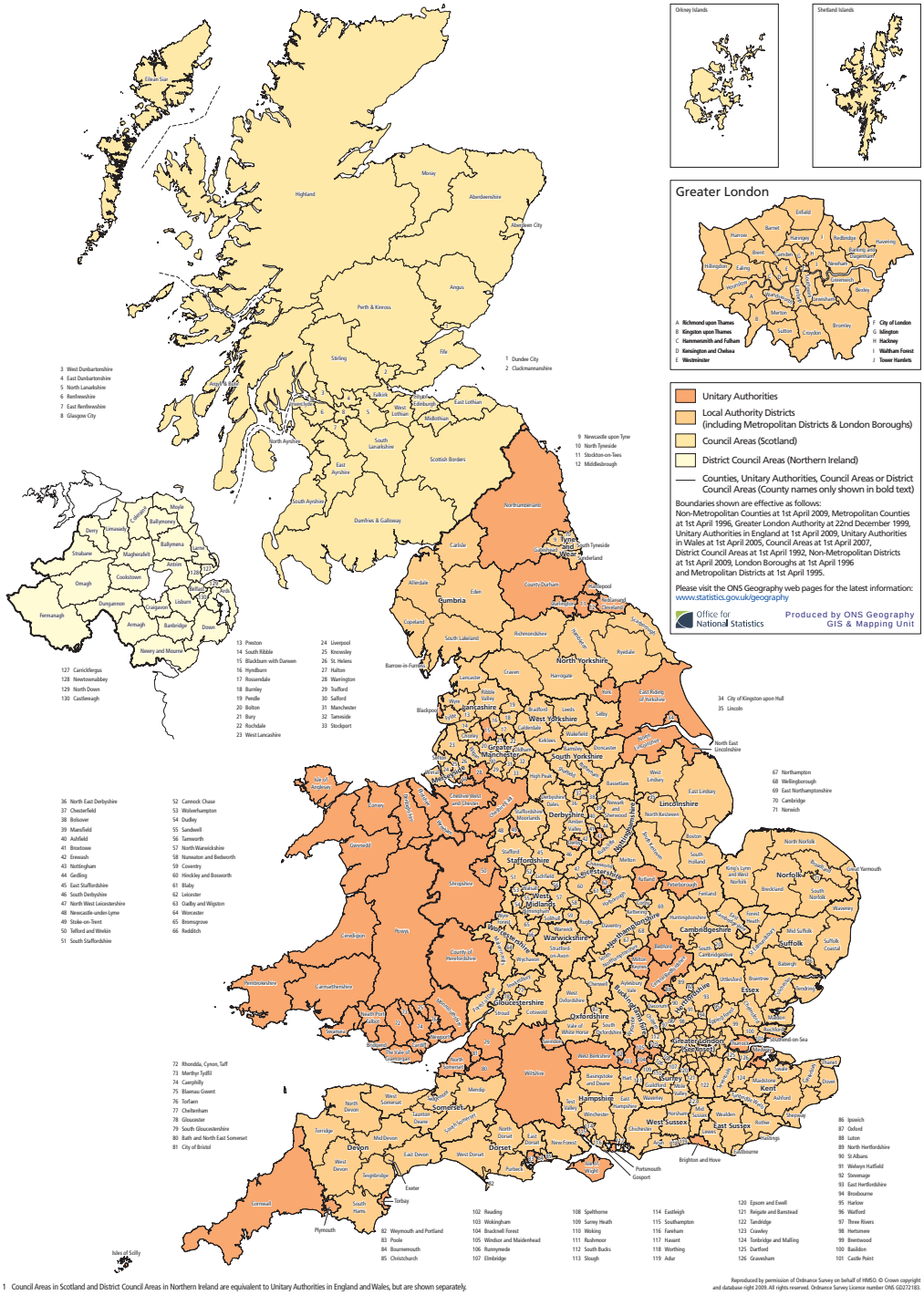


Figure 8: Local Authority Districts in the UK [2]